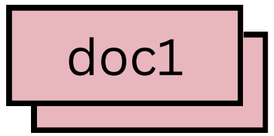
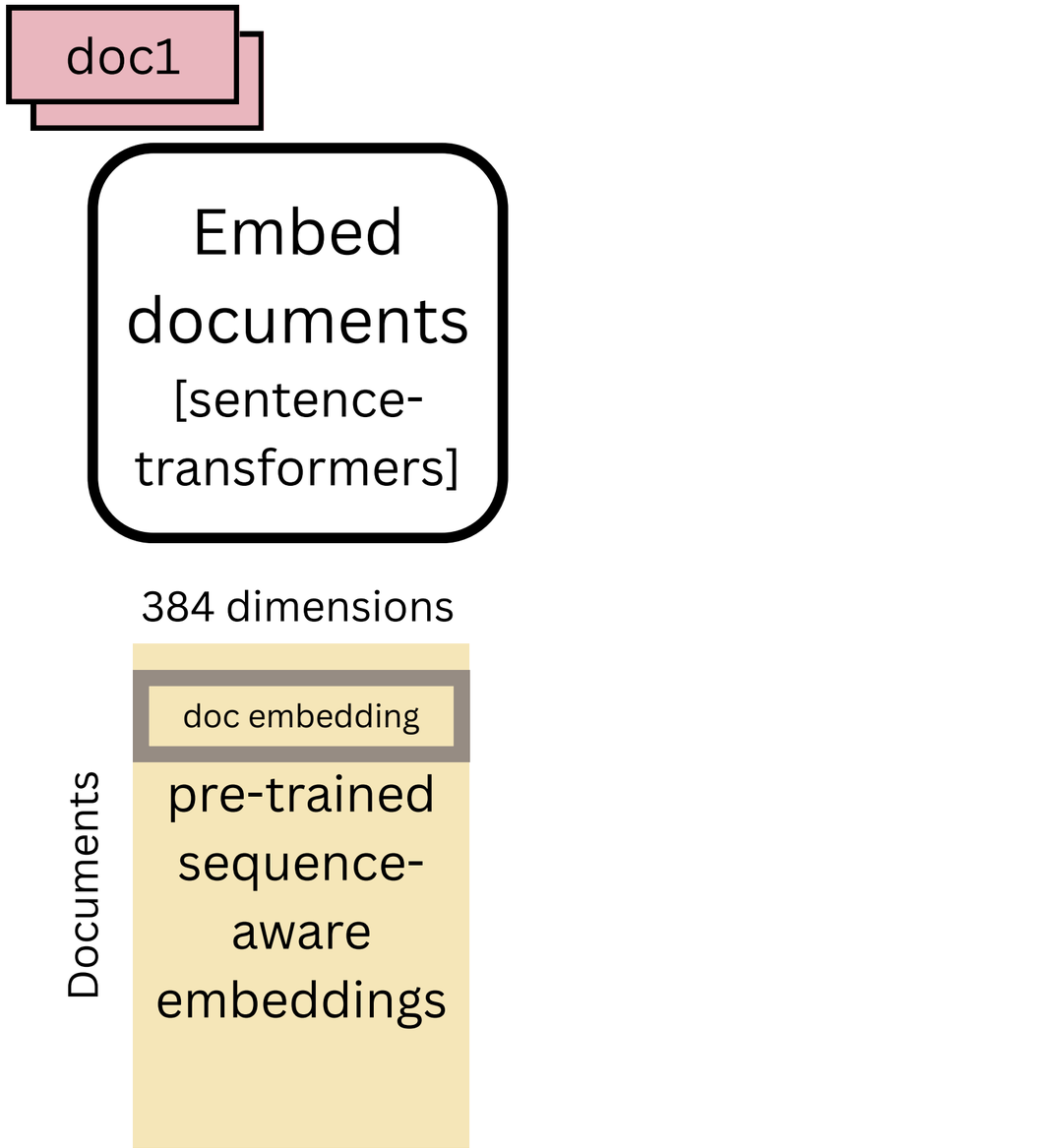
Natural Language Processing
Lab 7: Topic Models continued

Lonny Chen: l.chen@hertie-school.org
GRAD-E1282, Fall 2026
14-15 October 2026

BERTopic



BERTopic


doc1
Embed documents
[sentence-transformers]
384 dimensions
doc embedding
pre-trained sequence-aware embeddings
Documents

BERTopic

doc1

Embed documents [sentence-transformers]

384 dimensions

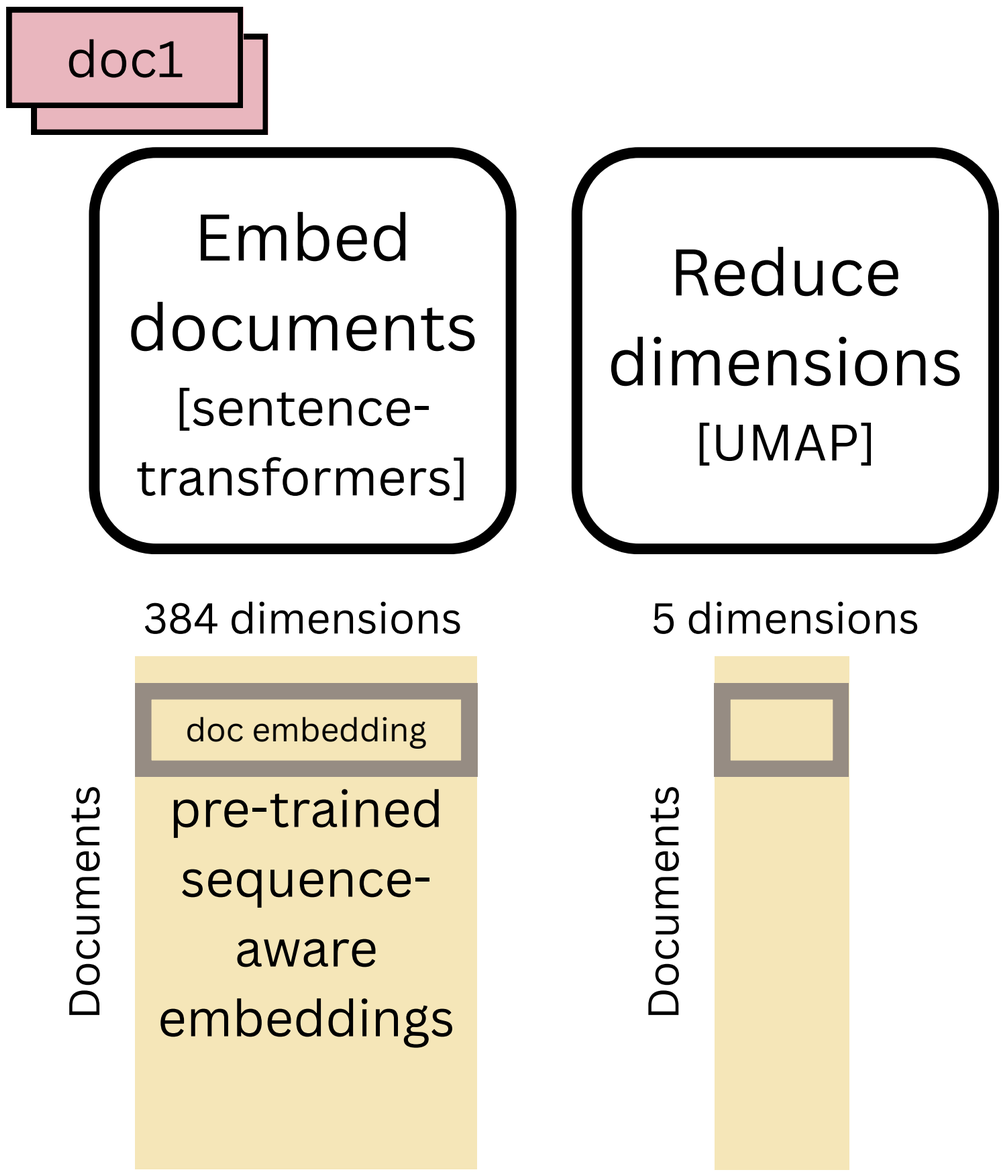
Documents

doc embedding

pre-trained sequence-aware embeddings

document	0	1	2	3	4	5	6	7	8	9
0	-0.028305	-0.049473	0.058456	-0.042718	0.061564	0.047414	0.017294	0.005074	-0.099917	0.042212
1	-0.042832	-0.031016	0.021979	-0.028294	0.063540	0.003272	0.038178	0.021101	-0.049943	0.046691
2	-0.013891	0.069305	-0.052530	-0.074148	0.042245	0.035878	0.013128	0.020967	0.020933	0.079212
3	-0.043826	-0.030694	0.010418	-0.041886	0.028059	0.006537	-0.004983	0.036859	0.010255	-0.018379
4	0.011813	-0.034563	0.006444	0.002248	0.082585	0.055026	0.011531	0.004346	-0.014189	0.041501

BERTopic


doc1
Embed documents
[sentence-transformers]
Reduce dimensions
[UMAP]
384 dimensions
5 dimensions
doc embedding
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Documents
Documents

BERTopic

doc1

Embed documents [sentence-transformers]

Reduce dimensions [UMAP]

Cluster docs into topics [HDBSCAN]

384 dimensions

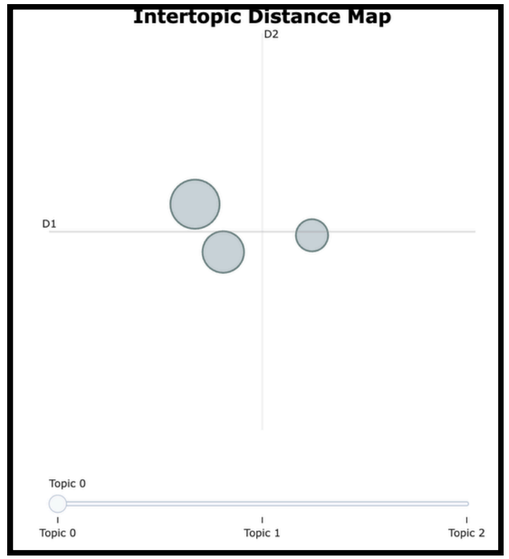
Documents

doc embedding

pre-trained sequence-aware embeddings

5 dimensions

Documents



ignore outliers (topic -1)

BERTopic

doc1

Embed documents [sentence-transformers]

Reduce dimensions [UMAP]

Cluster docs into topics [HDBSCAN]

Vectorize clustered docs [CountVect.]

Weigh Terms by cluster [cTF-IDF]

384 dimensions

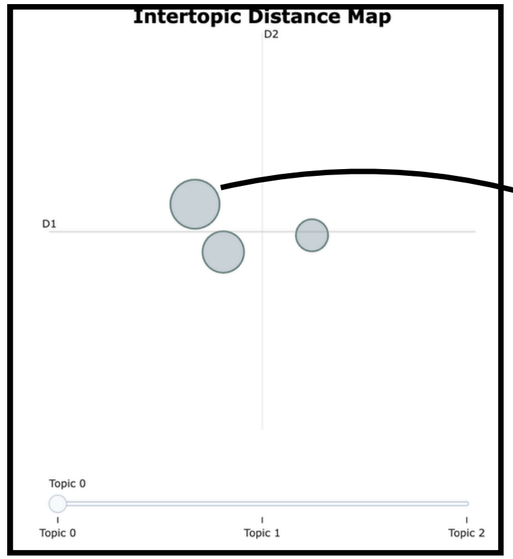
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doc embedding

pre-trained sequence-aware embeddings

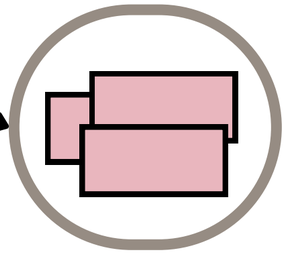
5 dimensions

Documents



ignore outliers (topic -1)

Topic n



BERTopic

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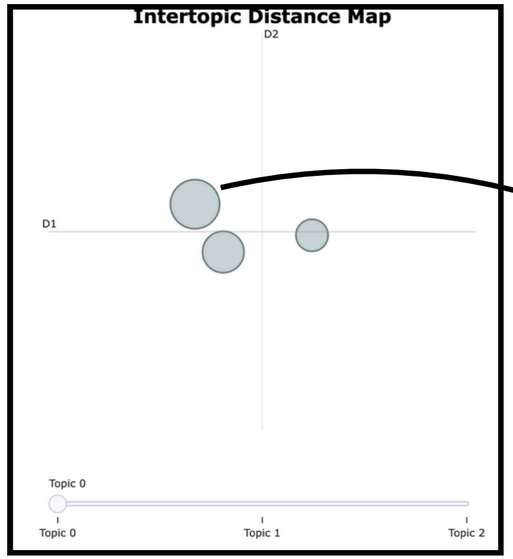
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pre-trained sequence-aware embeddings

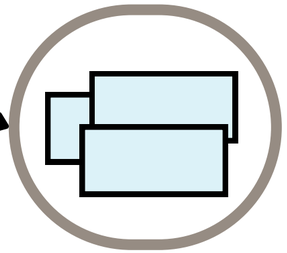
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Documents



ignore outliers (topic -1)

Topic n



BERTopic

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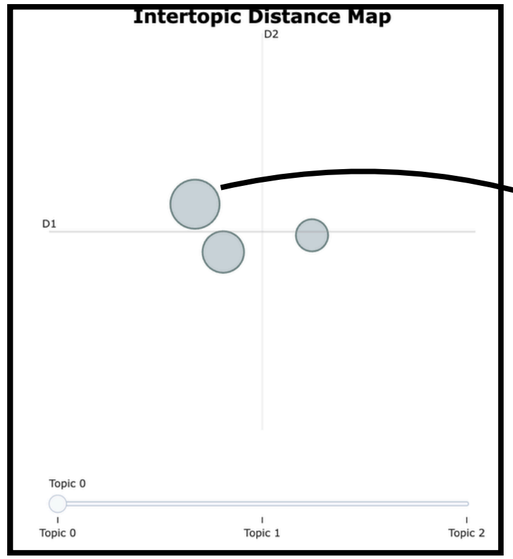
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Documents



ignore outliers (topic -1)

Topic n

Topics

Vocab terms

class TF-IDF weights

BERTopic

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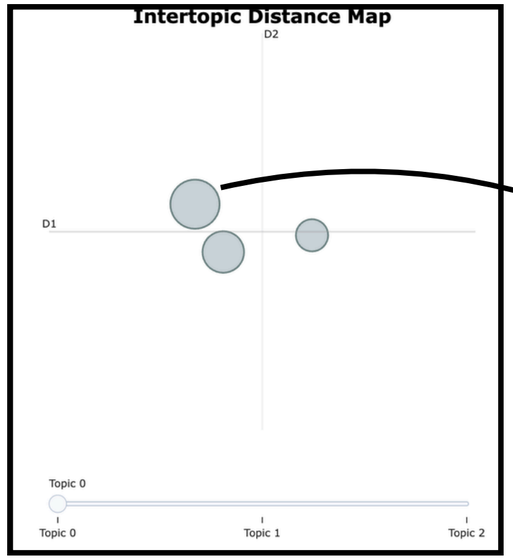
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pre-trained sequence-aware embeddings

5 dimensions

Documents



ignore outliers (topic -1)

Topic n

Vocab terms

class TF-IDF weights

Topics

Topic	aalberg	aaron	aart	ab	abacus
-1	0.000000	0.000000	0.000000	0.000000	0.000000
0	0.000000	0.001064	0.000000	0.001064	0.000000
1	0.000942	0.000000	0.000000	0.000000	0.000942
2	0.000000	0.000000	0.002456	0.000000	0.000000

BERTopic

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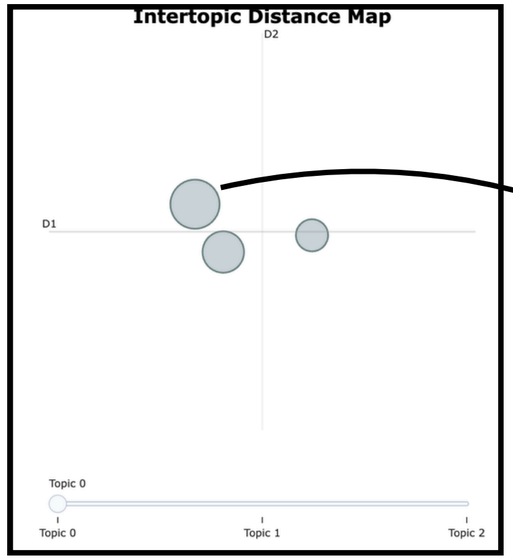
Documents

doc embedding

pre-trained sequence-aware embeddings

5 dimensions

Documents



ignore outliers (topic -1)

Topic n

Topics

Vocab terms

class TF-IDF weights

	Topic	Count	Name
0	-1	3	
1	0	7	
2	1	5	
3	2	3	

BERTopic - Reducing K Topics

doc1

Embed documents [sentence-transformers]

Reduce dimensions [UMAP]

Cluster docs into topics [HDBSCAN]

Vectorize clustered docs [CountVect.]

Weigh Terms by cluster [cTF-IDF]

384 dimensions

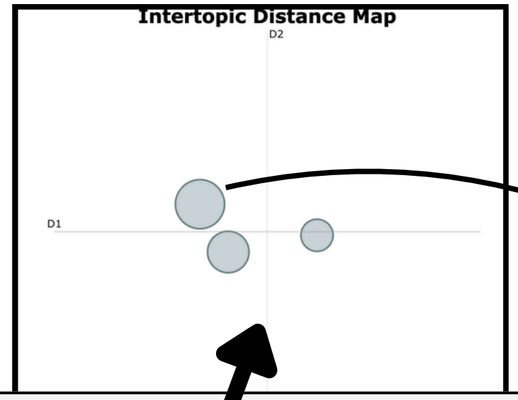
Documents

doc embedding

pre-trained sequence-aware embeddings

5 dimensions

Documents



Topic n

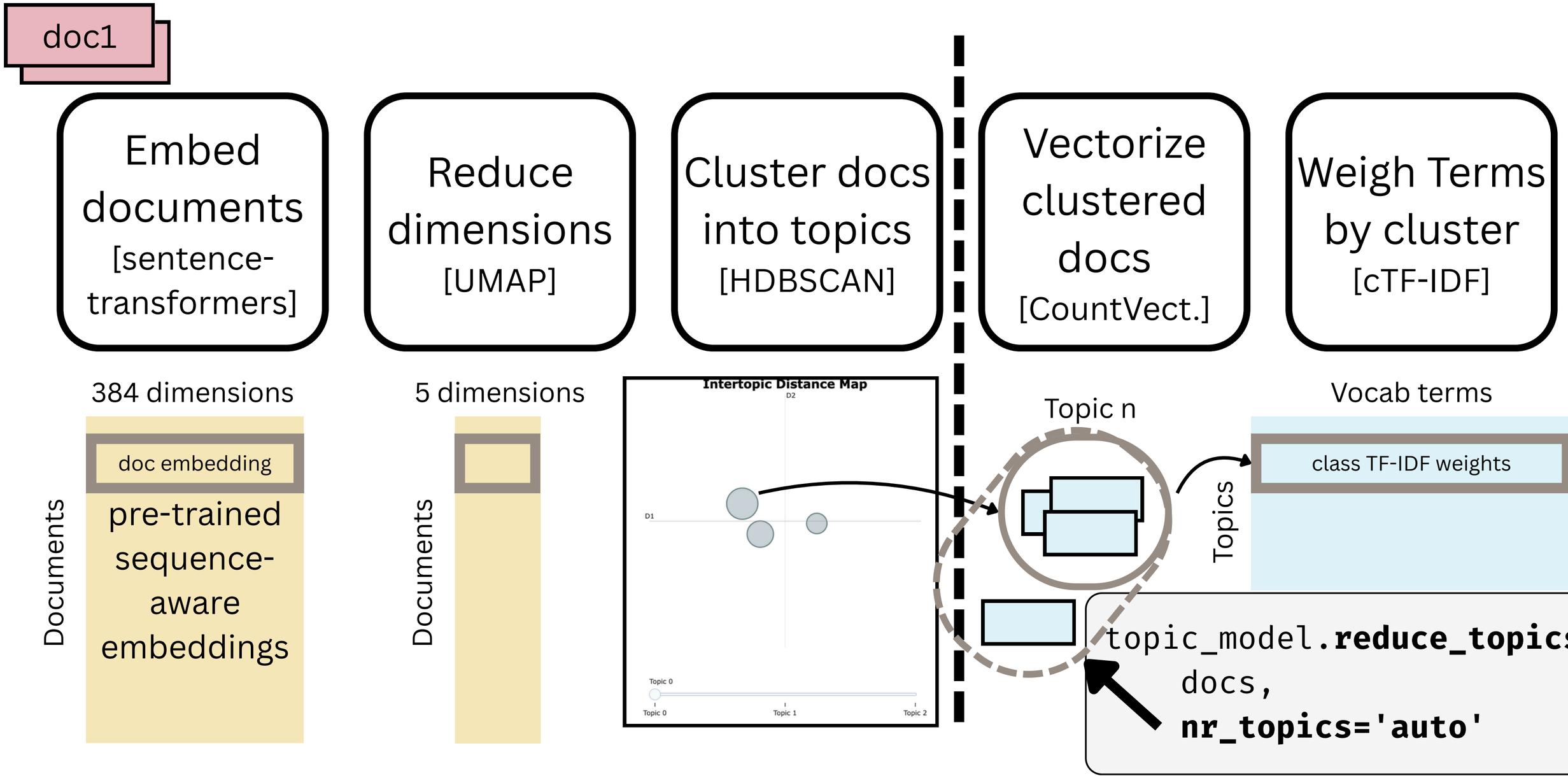
Topics

Vocab terms

class TF-IDF weights

```
topic_model = BERTopic(
    min_topic_size = 20 # default 10
```

BERTopic - Reducing K Topics



BERTopic - Reducing K Topics

doc1

Embed documents [sentence-transformers]

Reduce dimensions [UMAP]

Cluster docs into topics [HDBSCAN]

Vectorize clustered docs [CountVect.]

Weigh Terms by cluster [cTF-IDF]

384 dimensions

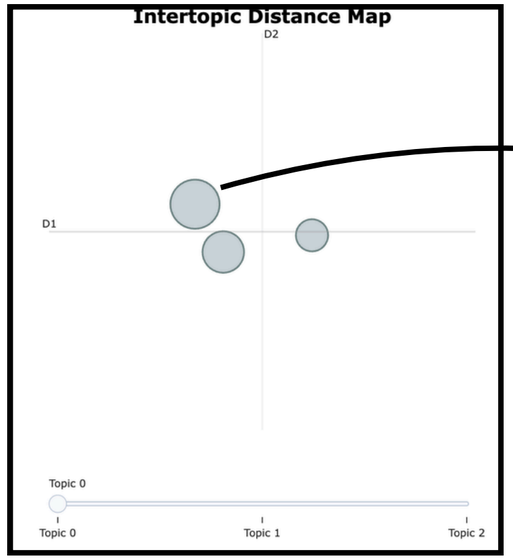
Documents

doc embedding

pre-trained sequence-aware embeddings

5 dimensions

Documents



Topic n

Topics

Vocab terms

class TF-IDF weights

```
topic_model.merge_topics(
    docs,
    [1, 2])
```

BERTopic - Reducing K Topics

doc1

Embed documents [sentence-transformers]

Reduce dimensions [UMAP]

Cluster docs into topics [HDBSCAN]

Vectorize clustered docs [CountVect.]

Weigh Terms by cluster [cTF-IDF]

384 dimensions

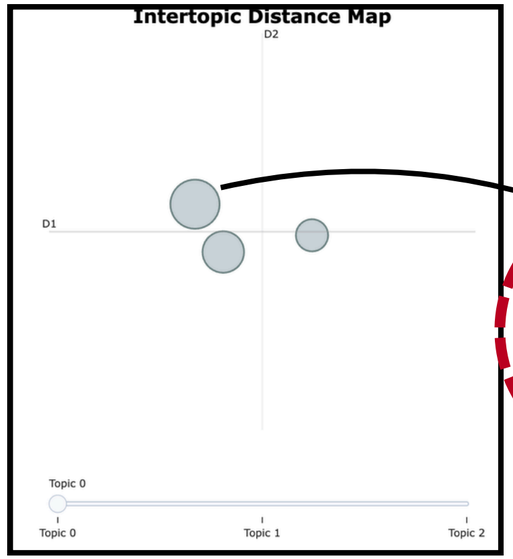
Documents

doc embedding

pre-trained sequence-aware embeddings

5 dimensions

Documents



Topic n

Topics

Vocab terms

class TF-IDF weights

```
topic_model.reduce_topics(
    docs,
    nr_topics=2
```

BERTopic - Reducing K Topics

Topic Reduction

BERTopic uses HDBSCAN for clustering the data and it cannot specify the number of clusters you would want. To a certain extent, this is an advantage, as we can trust HDBSCAN to be better in finding the number of clusters than we are. Instead, we can try to reduce the number of topics that have been created. Below, you will find three methods of doing so.

Warning

For all cases of topic reduction it is generally advised to create the number of topics you would first through the clustering algorithm. That tends to be the most stable technique and often gives you the best results. This also applies with algorithms that do not allow you to select the number of topics beforehands, like HDBSCAN where you can make sure of the `min_cluster_size` parameter to control the number of topics. Therefore, it is **highly** advised to not use `nr_topics` before you have attempted to control the number of topics through the clustering algorithm!

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Source: BERTopic Topic Reduction

BERTopic - Increasing K Topics

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Cluster docs into topics [HDBSCAN]

Vectorize clustered docs [CountVect.]

Weigh Terms by cluster [cTF-IDF]

384 dimensions

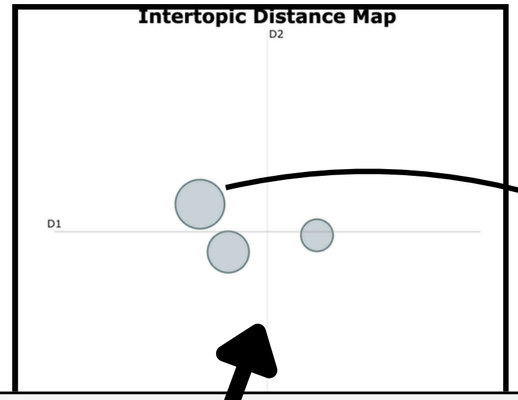
Documents

doc embedding

pre-trained sequence-aware embeddings

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Documents



Topic n

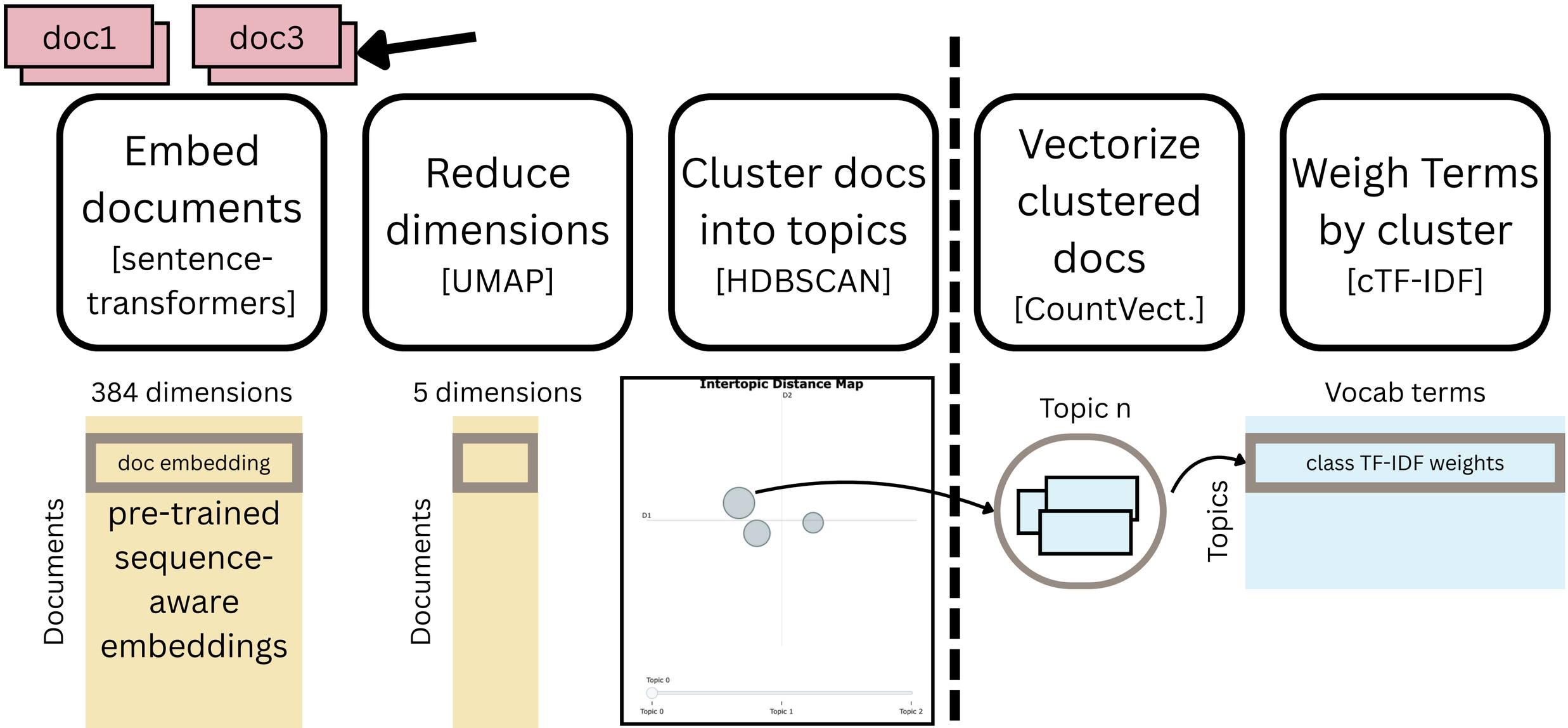
Topics

Vocab terms

class TF-IDF weights

```
topic_model = BERTopic(
    min_topic_size = 5 # default 10
```

BERTopic - Increasing K Topics



Coherence

```
'The heatwave pushed hospital admissions up as energy prices hit
records and drought hurt the harvest.'
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Coherence

```
'The heatwave pushed hospital admissions up as energy prices hit
records and drought hurt the harvest.'
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

Coherence measures word similarity WITHIN a topic compared to a reference corpus.

Coherence

['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’, ‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Coherence

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave)
Pr(hospital)
Pr(admissions)

Pr(heatwave, hospital)
Pr(heatwave, admissions)
Pr(hospital, admissions)

Coherence

window_size = 5

```
['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices',
'hit', 'records', 'drought', 'hurt, 'harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave) = ***1 / 1***
Pr(hospital) = ***1 / 1***
Pr(admissions) = ***1 / 1***

Pr(heatwave, hospital). = ***1 / 1***
Pr(heatwave, admissions) = ***1 / 1***
Pr(hospital, admissions) = ***1 / 1***

Coherence

window_size = 5

['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’, ‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave) = ***1 / 2***
Pr(hospital) = ***2 / 2***
Pr(admissions) = ***2 / 2***

Pr(heatwave, hospital). = ***1 / 2***
Pr(heatwave, admissions) = ***1 / 2***
Pr(hospital, admissions) = ***2 / 2***

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave) = ***1 / 3***
Pr(hospital) = ***3 / 3***
Pr(admissions) = ***3 / 3***

Pr(heatwave, hospital). = ***1 / 3***
Pr(heatwave, admissions) = ***1 / 3***
Pr(hospital, admissions) = ***3 / 3***

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt’, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave) = ***1 / 4***
Pr(hospital) = ***3 / 4***
Pr(admissions) = ***4 / 4***

Pr(heatwave, hospital). = ***1 / 4***
Pr(heatwave, admissions) = ***1 / 4***
Pr(hospital, admissions) = ***3 / 4***

Coherence

window_size = 5

['heatwave', ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’, ‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave) = ***1 / 5***
Pr(hospital) = ***3 / 5***
Pr(admissions) = ***4 / 5***

Pr(heatwave, hospital). = ***1 / 5***
Pr(heatwave, admissions) = ***1 / 5***
Pr(hospital, admissions) = ***3 / 5***

Coherence

window_size = 5

```
['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices',
'hit', 'records', 'drought', 'hurt', 'harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave) = ***1 / 6***
Pr(hospital) = ***3 / 6***
Pr(admissions) = ***4 / 6***

Pr(heatwave, hospital). = ***1 / 6***
Pr(heatwave, admissions) = ***1 / 6***
Pr(hospital, admissions) = ***3 / 6***

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt’, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave) = ***1 / 7***
Pr(hospital) = ***3 / 7***
Pr(admissions) = ***4 / 7***

Pr(heatwave, hospital). = ***1 / 7***
Pr(heatwave, admissions) = ***1 / 7***
Pr(hospital, admissions) = ***3 / 7***

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{PMI}(w_i, w_j) = \ln\left(\frac{P(w_i, w_j)}{P(w_i)P(w_j)}\right)$$

PMI(heatwave, hospital) = ln ((1 / 7) / (1/7)(3/7)) = 0.8473

PMI(heatwave, admissions) = ln ((1 / 7) / (1/7)(4/7)) = 0.5596

PMI(hospital, admissions) = ln ((3 / 7) / (3/7)(4/7)) = 0.5596

Coherence

window_size = 5

['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices', 'hit', 'records', 'drought', 'hurt, 'harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

NPMI(heatwave, hospital) = ln ((1 / 7) / (1/7)(3/7)) / -ln(1 / 7) = 0.4351

NPMI(heatwave, admissions) = ln ((1 / 7) / (1/7)(4/7)) / -ln(1 / 7) = 0.2871

NPMI(hospital, admissions) = ln ((3 / 7) / (3/7)(4/7)) / -ln(3 / 7) = 0.6216

Coherence

window_size = 5

['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices', 'hit', 'records', 'drought', 'hurt, 'harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

NPMI(topic_0) = AVERAGE(NPMI(w_i, w_j)) ≈ 0.4479

Coherence

window_size = 5

['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices', 'hit', 'records', 'drought', 'hurt, 'harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

NPMI(topic_0) = AVERAGE(NPMI(w_i, w_j)) ≈ 0.4479
NPMI(topic_1) = AVERAGE(NPMI(w_i, w_j)) ≈ -0.4409

Coherence

window_size = 5

['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices', 'hit', 'records', 'drought', 'hurt, 'harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

Coherence_NPMI = AVERAGE(NPMI(topic_k)) ≈ 0.01

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'admissions', 'harvest']
topic_1 = ['hospital', 'records', 'hurt']
```

```
topic_0 = ['heatwave', 'pushed', 'admissions']
topic_1 = ['admissions', 'energy', 'prices']
```

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'admissions', 'harvest']
topic_1 = ['hospital', 'records', 'hurt']
```

Coherence_NPMI ≈ -0.48

```
topic_0 = ['heatwave', 'pushed', 'admissions']
topic_1 = ['admissions', 'energy', 'prices']
```

Coherence_NPMI ≈ 0.37

Diversity

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Diversity measures word uniqueness COMPARED TO other topics

Diversity

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Diversity measures word uniqueness COMPARED TO other topics

Diversity = Unique words / Total words

Diversity

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’, ‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Diversity measures word uniqueness COMPARED TO other topics

Diversity = 5 /6 = 0.83

Diversity

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'admissions', 'harvest']
topic_1 = ['hospital', 'records', 'hurt']
```

Diversity = 1.0

```
topic_0 = ['heatwave', 'pushed', 'admissions']
topic_1 = ['admissions', 'energy', 'prices']
```

Diversity = 0.83

Mid-semester check-in



Next Week: Neural Networks!

